

LScAD: A Large–Small Model Collaboration Framework for Unsupervised Industrial Anomaly Detection

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Abstract—Currently, in the field of anomaly detection, most existing methods rely on small models that focus on specific industrial scenarios which exhibit strong task orientation but lack generalization capabilities. Segment anything model (SAM), a vision foundation model designed for semantic segmentation tasks, has demonstrated remarkable performance in the natural scene segmentation. However, SAM lacks domain-specific knowledge of industrial defects and relies on an interactive inference framework, which restricts its application in industrial anomaly detection. In this article, we propose a novel framework: A large–small model collaboration framework for unsupervised industrial anomaly detection (LScAD), aiming to use task-oriented small models to guide SAM for precise anomaly localization. Specifically, the small model generates the initial guidance information, which serves as the input for a multimodal prompt module. This module consists of two modalities: image and text prompt. These prompts are then used to guide SAM for accurate anomaly segmentation. Moreover, we design a dual-branch adapter to enhance SAM’s domain-specific capability through a color-domain branch and a frequency-domain branch, aiming to improve its performance in anomaly detection tasks. Extensive experiments on the MVTec AD benchmark and other real-world industrial datasets demonstrate that our method achieves state-of-the-art performance, with an image-level area under the receiver operating characteristic curve (AUROC) of 99.6%, pixel-level AUROC of 98.4%, and average precision (AP)

of 74.1% on the MVTec AD dataset. Our proposed method can also effortlessly adapt to multiclass anomaly detection without any modifications and achieve remarkable performance. Our code is available at: <https://github.com/qsc1103/LScAD>

Index Terms—Anomaly detection, large–small model collaboration, segment anything model (SAM), unsupervised learning.

I. INTRODUCTION

INDUSTRIAL defect detection based on supervised deep learning is a critical technology widely used in fields, such as mechanical manufacturing [1], consumer electronics [2], and bridge inspection [3]. With the advancement of computer vision technologies, supervised learning methods have shown impressive performance in industrial defect detection. However, these approaches heavily rely on large-scale, pixel-level annotated datasets, which are extremely costly and time-consuming to obtain in real-world manufacturing environments [4]. Moreover, the diversity of defects, which includes variations in shape, scale, and texture, makes it difficult to collect sufficient annotated samples to cover all possible types of anomalies [5]. These limitations have motivated a shift to unsupervised paradigms, where models are trained only on normal samples and are expected to identify deviations during inference [6].

Currently, most unsupervised anomaly detection methods rely on specific task-oriented small models, which are typically lightweight convolutional neural networks (CNNs) with a limited number of parameters and without pretraining on large-scale datasets [5]. Existing small-model-based approaches can be roughly divided into three categories: feature embedding-based methods [7], [8], [9], which aim to learn unique feature embeddings and detect anomalies based on the distance between learned features and input features in the feature space; reconstruction-based methods [10], [11], [12], [13], which rely on reconstruction errors between abnormal and normal regions to detect anomalies; and anomaly synthesis-based methods [14], [15], [16], [17], which introduce the synthetic anomaly information to detect anomalies in a self-supervised paradigm. As shown in Fig. 1(a), current task-specific small models usually introduce refinement modules represented by U-Net [14] or simple multilayer perceptrons [15] to further improve the performance of anomaly localization. However,

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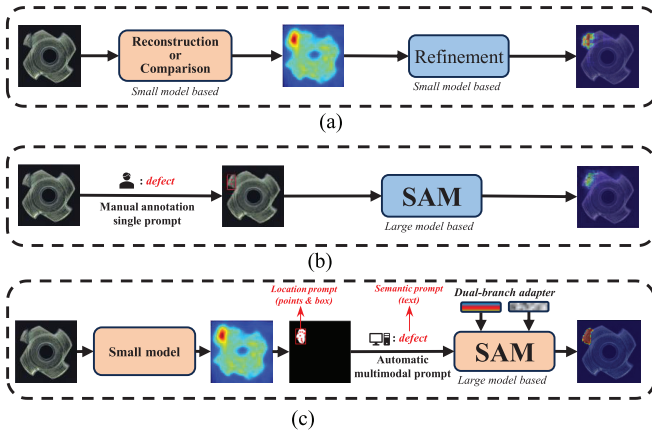


Fig. 1. Comparison of different frameworks for anomaly detection. (a) Specific task-oriented small model. (b) Manual annotation single-prompt-based SAM. (c) Automatic multimodal prompt-based SAM with dual-branch adapter.

due to the limited number of parameters, small models inherently lack the capacity to extract and represent high-level semantic features of anomalies. This limitation significantly weakens their ability to distinguish subtle defect patterns from normal textures, especially when dealing with complex backgrounds or low-contrast variations. Furthermore, in the absence of pretraining on large-scale and diverse datasets, these models fail to acquire generalizable knowledge, which is essential for adapting to unseen scenarios and variations in lighting, scale, noise, or structural patterns. As a result, small models often suffer from poor generalization and limited robustness, making it difficult to accurately detect and localize defects in real-world industrial environments [5].

Recently, with the rapid rise of large-scale pretrained techniques, meta-AI research proposed an interactive vision foundation model for image segmentation called the segment anything model (SAM) [18]. It is trained on a billion-scale segmentation dataset, achieving impressive zero-shot segmentation performance in natural scenes. With a large number of parameters and a powerful feature extraction backbone, it possesses strong representational capacity, enabling remarkable performance across various downstream vision tasks [5]. However, applying SAM to industrial anomaly detection presents several significant challenges. First, SAM's mask prediction is inherently interactive and relies on manually provided prompts, as illustrated in Fig. 1(b), which limits its applicability in fully automated industrial inspection scenarios [19]. Second, the data distribution of SAM's pretraining corpus differs substantially from that of industrial environments, lacking domain-specific characteristics that are essential for reliable generalization to anomaly detection tasks in manufacturing [20]. Finally, existing SAM-based anomaly detection approaches typically utilize SAM either as a standalone segmentation module [21] or as a feature extractor backbone [22], [23], without fully leveraging its potential for understanding anomaly semantics. Therefore, directly applying large pretrained models such as SAM to industrial anomaly detection is not an appropriate choice.

In order to combine the generative capabilities of large models and the specific task orientation of small models, a novel large–small model collaboration framework for unsupervised industrial anomaly detection (LScAD) is proposed, as shown in Fig. 1(c). Specifically, different from previous SAM-based anomaly detection methods, a small model built on a teacher–student network is designed to address the limitations of SAM's interactive working framework. During training, since only normal samples are provided, a self-supervised strategy based on synthetic anomaly images is employed to train the multimodal prompt module. Unlike conventional single-prompt methods, the proposed multimodal prompt module incorporates both location and semantic prompts, which effectively leverage the preliminary guidance signals generated by the small model to enhance the anomaly localization task [24]. This design thereby effectively guides SAM to perform more refined anomaly segmentation tasks. Furthermore, existing large models often lack adaptability to industrial scenarios, to address this issue, our method introduces a dual-branch adapter structure composed of a frequency-domain adapter and a color-domain adapter. These adapters enrich the feature representation by capturing high-frequency textures and color variations, ensuring the large model is well-suited for industrial applications.

The architecture of our LScAD is feasible because it combines the efficiency of small models with the expressiveness of large models, while addressing domain gap and representation limitations through prompt guidance and feature adaptation.

In summary, our contributions are as follows.

- 1) A novel framework LScAD is proposed to guide the general segmentation ability of SAM with the task-oriented capability of the small model, establishing a collaborative working approach for industrial anomaly detection.
- 2) Combining the small model with the multimodal prompt module, we have not only resolved the interactive issue but also generated an efficient prompt for SAM. Furthermore, a dual-branch adapter is designed to inject color and frequency-domain knowledge into SAM for industrial anomaly detection.
- 3) Extensive experiments are conducted on the MVTec AD, WFDD, and BTAD datasets under the single-class/multiclass case which demonstrate that LScAD achieves state-of-the-art performance in both anomaly detection and localization. Meanwhile, our method is successfully applied in a real-world vision system to detect chip trays.

II. RELATED WORKS

This section reviews related works on industrial anomaly detection and localization, which can be divided into two main categories: task-oriented small model-based methods and SAM-based methods.

A. Task-Oriented Small-Model-Based Methods

1) *Feature Embedding-Based Methods:* Feature embedding-based methods typically utilize pretrained

networks to extract semantic representations and project them into a lower dimensional latent space [25], where anomalies are identified by measuring the deviation of testing sample embeddings from the distribution of normal training embeddings. RD4AD [7] proposes the research paradigm of reverse distillation, which effectively alleviates the limitation of anomaly representation. PaDim [9] utilizes multivariate Gaussian distributions to obtain a probabilistic representation and evaluates anomaly scores via the Mahalanobis distance. Patchcore [8] uses a maximally representative memory bank of nominal patch features, which offers promising inference times and detection accuracy.

2) *Reconstruction-Based Methods*: The reconstruction-based methods rely on the strong assumption that normal regions can be reconstructed almost perfectly by the model, while abnormal regions cannot be reconstructed. DFR [10] proposes a multiscale regional feature generator and a lightweight convolutional autoencoder to detect and locate anomalies via fast feature reconstruction. Some methods set the condition that the abnormal pixels are not visible to the autoencoder and reconstruct the missing information through inpainting. Typically, RIAD [11] and Intra [26] use CNN and Transformer, respectively, to inpaint images. However, the generalization ability of the generated model is often underestimated, and the network can not only reconstruct normal regions but also generalize to abnormal regions, resulting in inaccurate reconstruction errors.

3) *Anomaly Synthesis-Based Methods*: The existing anomaly synthesis methods can be divided into two paradigms: image-level anomaly synthesis and feature-level anomaly synthesis. The anomaly synthesis strategy of DRAEM [14] creates binary masks using Perlin noise and fills them with various textures from auxiliary datasets. Tao et al. [27] adopted a random erasure method to generate more realistic defect images. These image-level synthesized anomalies are primarily utilized to train models in a self-supervised manner. In contrast to image-level anomaly synthesis, feature-level anomaly synthesis is typically conducted in high-dimensional feature spaces. SimpleNet [15] adds random Gaussian noise sampled from a normal distribution to each feature channel of normal samples. DSR [28] generates the anomalies at the feature level by sampling the learned quantized feature space, which allows a controlled generation of near-in-distribution anomalies.

While these methods have achieved promising detection performance, they are constrained by the ability of small models, making it difficult to achieve a perfect anomaly localization effect.

B. Sam-Based Methods

With the rapid rise of large-scale vision foundation models, the universal interactive segmentation model SAM has achieved remarkable performance in natural scenes. However, the bias in SAM's training data distribution limits its effectiveness in downstream tasks. To address the adaptation challenges of SAM across diverse scenarios, researchers have focused on efficient fine-tuning techniques, such as LoRA [29] and Adapter [30]. However, the current existing adapter

structures all originate from natural language processing tasks [31], and there is no adapter architecture specifically designed for the purpose of anomaly detection yet. Some studies have also conducted preliminary explorations of applying SAM to anomaly detection tasks. ClipSAM [21] leverages the powerful general capabilities of CLIP and SAM to perform two-stage zero-shot anomaly detection. STLM [23] designs a dual-stream lightweight model and utilizes SAM for knowledge guidance. UCAD [22] designs a structure-based contrastive learning module based on SAM to enhance the performance of prompt learning and continual anomaly detection. The above SAM-based methods do not incorporate domain knowledge of industrial anomalies, resulting in difficulty achieving satisfactory performance under the settings of diverse industrial anomaly detection scenarios.

III. METHOD

A. Overview

The overall pipeline of our method is illustrated in Fig. 2, which consists of three key components: small model, the multimodal prompt module, and SAM-based large model with a dual-branch adapter.

We adopt a single-stage training strategy for our proposed method, as shown in Fig. 3(a). To further elaborate on the internal structure and detailed functionality of each component, we refer to Fig. 2. During training, a normal sample I_N is used to train the small model, which is a lightweight teacher-student reverse knowledge distillation network. Since only normal samples are available for training, we design a self-supervised training approach to train the multimodal prompt module for generating location prompts and semantic prompt embedding. Specifically, anomaly synthesis strategy is employed to generate an anomalous image I_A and its corresponding mask M_A , so the location prompt is obtained by points and boxes automatically annotated from the known synthetic mask M_A . The semantic prompt embedding is generated by fusing and aligning the anomalous sample I_A and its corresponding text t_p through the cross-modal model BEiT3 [32] and an MLP projection layer. The text serves as a semantic prior to guide the large model in interpreting the visual content from a multimodal perspective. This design takes advantage of the vision-language alignment learned during pretraining. The location prompt and semantic prompt, derived from both the mask M_A and text t_p , are subsequently input into SAM's prompt encoder. This encoder serves to guide the SAM-based large model to achieve better anomaly detection results. In order to facilitate the migration of data distribution from natural scenes to industrial scenarios, a dual-branch adapter is delicately designed and then integrated into the original SAM. The process of embedding the adapter into SAM is as follows: the high-frequency components I_A^h of the anomalous image I_A are first extracted by a Butterworth high-pass filter (BHPF). Then, I_A^h is input into the proposed frequency-domain adapter, where it is transformed into frequency-domain embeddings and embedded before each Transformer block. Meanwhile, the color-domain adapter is embedded in the main and residual branches of each Transformer block in the SAM image encoder.

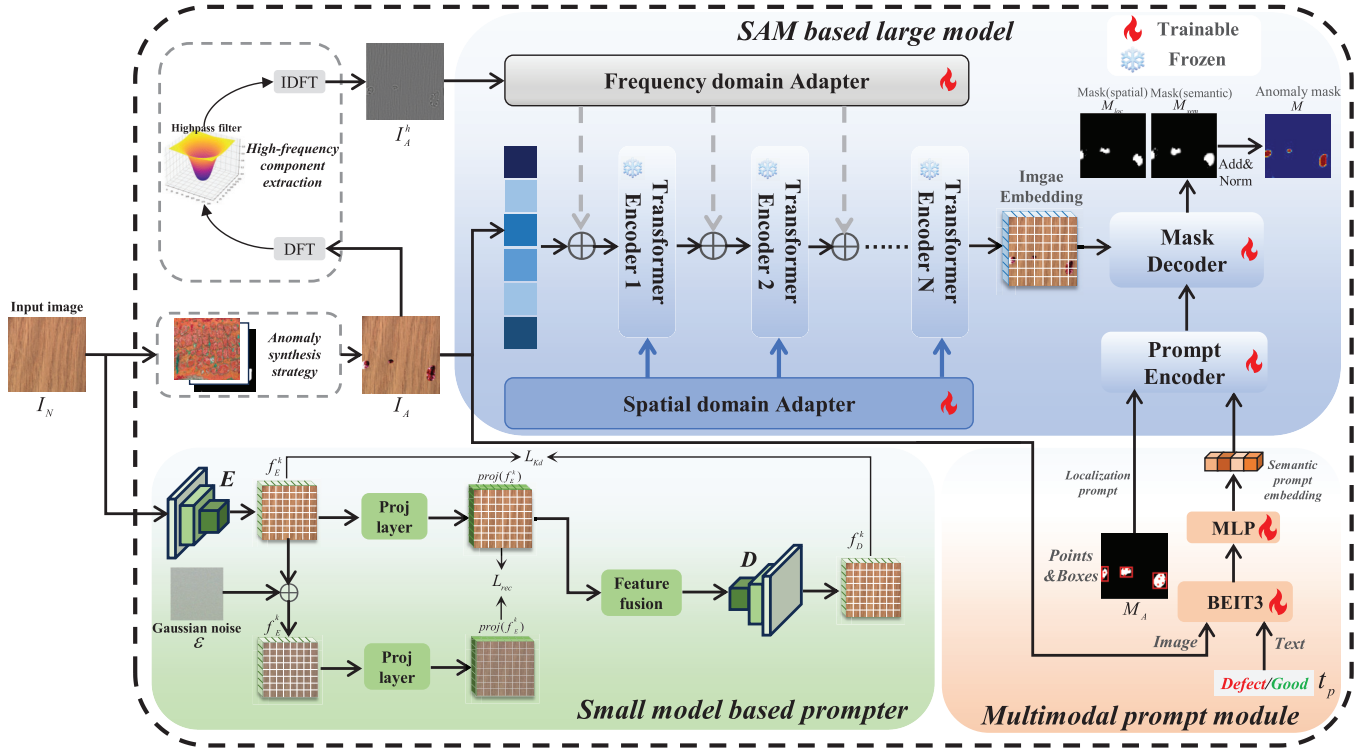


Fig. 2. Overview of our proposed LScAD framework.

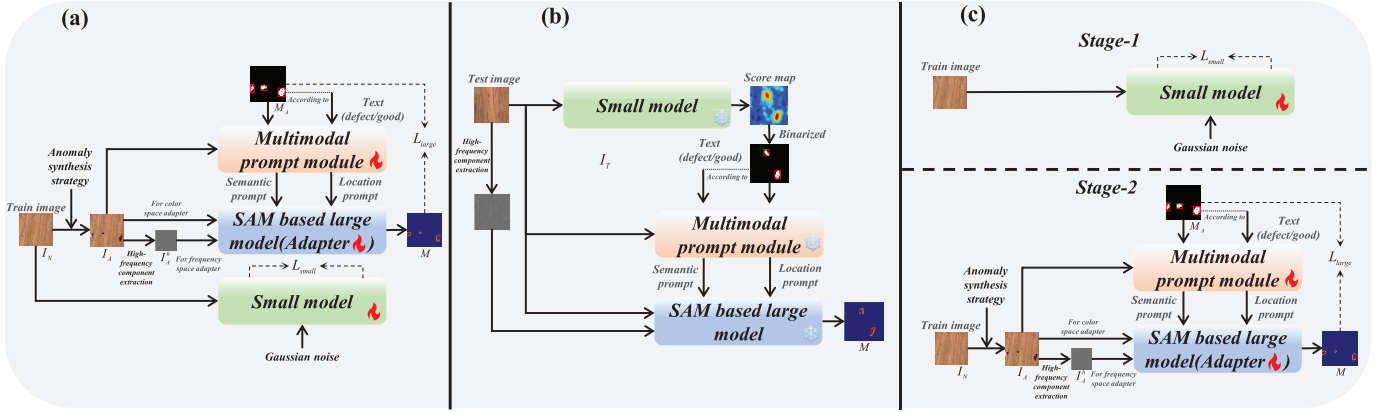


Fig. 3. Training and inference strategies. (a) Single-stage training strategy. (b) Inference strategy. (c) Two-stage training strategy.

During inference, the pipeline is shown in Fig. 3(b). The input image I_T is first fed into the small model. The score map of the multiscale feature between the output of encoder E and decoder D in the small model is then binarized and used as the initial guidance information. Based on the initial guidance information, the automatically annotated points, boxes, and its corresponding text are subsequently input into the multimodal prompt module to generate the location prompt and semantic prompt embedding. Finally, image I_T and its high-frequency component I_T^h are fed into the image encoder and the frequency adapter of SAM, respectively. Based on the above two types of prompts, SAM predicts the corresponding masks M_{loc} and M_{sem} for I_T . The final anomaly detection result M is obtained by adding and normalizing M_{loc} and M_{sem} , as shown in Fig. 2.

TABLE I
ARCHITECTURE OF THE TEACHER ENCODER

Stage	Operation	Output (Size × Channels)
Input	RGB Image	224 × 224 × 3
Conv1	7×7 Conv, stride 2 + BN + ReLU	112 × 112 × 64
Pool	3×3 MaxPool, stride 2	56 × 56 × 64
Res2	3 × Bottleneck Blocks	56 × 56 × 256
Res3	4 × Bottleneck Blocks	28 × 28 × 512
Res4	6 × Bottleneck Blocks	14 × 14 × 1024
Res5	3 × Bottleneck Blocks	7 × 7 × 2048

B. Small Model

1) *Architecture*: To address the limitations of SAM's interactive segmentation, a small model is proposed, as shown in

the green background at the bottom left of Fig. 2. Unlike other methods that require either complex anomaly synthesis strategies [14] or substantial storage to maintain external memory banks [8], we design the small model based on the reverse knowledge distillation paradigm, which has been proven to be an efficient and effective approach for anomaly detection [7]. The teacher encoder E is employed to extract comprehensive feature representations of the input image I_N , as detailed in Table I. WideResNet [33] pretrained on ImageNet serves as E to extract multiscale features f_E^k , where k represents the output of the k^{th} block of the teacher encoder. In addition, simple Gaussian noise ϵ is added to f_E^k , producing noisy feature representations $\tilde{f}_E^k$ to enhance the model's robustness. Subsequently, both f_E^k and $\tilde{f}_E^k$ undergo dimensionality reduction via a projection layer, and a bottleneck-based feature fusion module is then employed to align the multiscale features of the projected f_E^k . Finally, the fused features f_E^{fuse} are fed into the student decoder D to generate f_D^k , restoring them to the same dimension as the features of the teacher encoder. During training, the student decoder is trained to approximate the output of the teacher encoder as closely as possible, due to the representational capabilities between the teacher encoder and the student decoder, the cosine similarity between f_E^k and f_D^k is used to locate anomalous regions. Therefore, the small model is utilized to generate the initial guidance information for the following multimodal prompt module during inference.

2) *Loss Function*: The loss function of the small model includes the knowledge distillation loss [7] between f_E^k and f_D^k , as well as the cosine reconstruction loss [34] between $\tilde{f}_E^k$ and f_E^k , which can be formulated as follows:

$$\mathcal{L}_{\text{Kd}} = 1 - \sum_{k=1}^K \frac{(f_E^k)^\top \cdot (f_D^k)}{\|f_E^k\| \|f_D^k\|} \quad (1a)$$

$$\mathcal{L}_{\text{Rec}} = \frac{1}{K} \sum_{k=1}^K (1 - \cos(\text{Proj}(f_E^k), \text{Proj}(\tilde{f}_E^k))) \quad (1b)$$

$$\mathcal{L}_{\text{small}} = \mathcal{L}_{\text{Kd}} + \alpha \mathcal{L}_{\text{Rec}} \quad (1c)$$

where α is the hyperparameter that balances the loss weight in the small model. The optimal value of α is empirically determined to be 0.4 based on the ablation study in Section IV-E7.

C. Multimodal Prompt Module

As shown in the red background at the bottom right of Fig. 2, we design a multimodal prompt module that generates both location and semantic prompts based on the initial guidance from the small model, which are used to guide the large model to more accurately focus on the critical regions relevant to the anomaly detection task [35]. In this module, an anomaly synthesis strategy is adopted to generate pseudo-anomalies, and the corresponding masks of these pseudo-anomalies are used as initial guidance information to train the multimodal prompt module. However, during inference, the initial guidance information is derived from the binarized score map output by the small model.

Algorithm 1 Anomaly Synthesis Strategy

Input: Normal image I_N ; Anomaly source image A ; Perlin noise generator P ; Image size H and W ; Noise resolution r_x and r_y ; Blending factor γ ; Binarization threshold λ .

Output: Synthetic anomaly image I_A and corresponding mask M_A

- 1: $M_p \leftarrow \text{where}(P(H, W, r_x, r_y) > \lambda, 1, 0)$
- 2: $M_f \leftarrow \text{Foreground}(I_N)$
- 3: $\overline{M_A} \leftarrow M_p \cap M_f$
- 4: $\overline{M_A} \leftarrow 1 - M_A$
- 5: $I_A \leftarrow \gamma(M_A \odot A) + (1 - \gamma)(M_A \odot I_N) + \overline{M_A} \odot I_N$
- 6: **return** I_A, M_A

1) *Anomaly Synthesis Strategy*: We use the Perlin noise generator [14] to generate global anomaly masks M_p . Specifically, to ensure that the generated anomaly regions are located within the foreground of interest, a foreground segmentation binarization method based on the adaptive thresholding is proposed to obtain the foreground mask M_f . In practice, we apply Otsu or Triangle thresholding on grayscale images depending on the object subclass, followed by morphological closing and opening using a 10×10 square kernel to refine the mask [36]. The intersection of M_p and M_f is the final anomaly mask M_A . The whole process of anomaly synthesis strategy is shown in Algorithm 1. Following the same settings as [14], we set r_x and r_y to 4, the blending factor γ is randomly sampled from the range $[0, 1)$, and the binarization threshold λ is fixed to 0.5, resulting in a binary mask where values above the threshold are assigned 1 to indicate anomaly regions, and values below are assigned 0 as background.

2) *Location Prompt*: The location prompt includes the point and box coordinate information of the anomalous regions.

During training, the synthetic binary masks are generated, and ten random points are automatically annotated within the anomalous regions, with these points distributed at different locations of the defect. In addition, the smallest external rectangle that encompasses all disconnected regions is computed and used as the bounding box for the location prompt. This ensures that the location prompt provides comprehensive spatial information during the training phase, helping the model to identify and localize the anomalous regions.

In the inference phase, the score map generated by the small model is first binarized, converting the image into a mask-like representation. Then, this binarized mask is again automatically annotated with ten random points, similar to the process in the training phase. However, a key difference in the inference phase is that the smallest external rectangle of the largest connected region in the binarized mask is used as the bounding box for the location prompt. This ensures that the model focuses on the most prominent anomalous region within the image, thereby improving the accuracy of the localization process. The precise bounding box definition helps the model focus on the critical areas of the defect.

3) *Semantic Prompt*: For the semantic prompt, we first design a pair of opposing generic texts, “defect/good,” which correspond to the semantic information of anomalous and

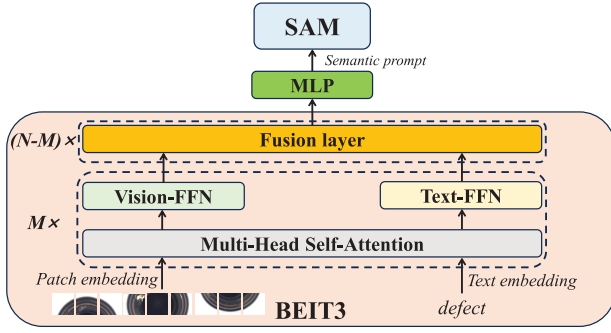


Fig. 4. Generation pipeline of the semantic prompt.

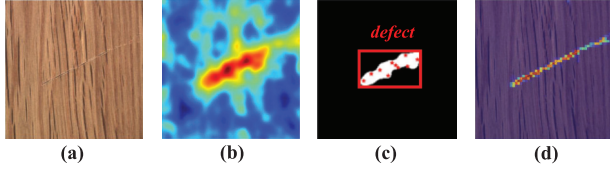


Fig. 5. Results of LScAD at different stages. (a) Testing image. (b) Output of the small model. (c) Location prompt (points and box) and semantic prompt (text). (d) Output of SAM. (b) and (d) are heatmaps, where red indicates regions with a higher likelihood of being anomalous, and blue represents normal regions.

normal samples, respectively. When the initial guidance information does not contain a defect, the text corresponds to “good”; otherwise, it corresponds to “defect.” This binary semantic labeling provides clear targets for subsequent model training.

To effectively integrate the textual and visual features of anomaly information, we employ the BEIT3 [32] multimodal encoder. BEIT3 leverages the powerful ability to jointly embed image and text, enabling fine-grained alignment in cross-modal learning tasks. Specifically, as shown in Fig. 4, the input image is first divided into nonoverlapping patches and projected into a sequence of visual tokens, while the input text prompt (e.g., “defect” or “good”) is tokenized and embedded into a sequence of textual tokens. These two token sequences are then concatenated and passed through a shared stack of multihead self-attention (MHSA) layers to enable cross-modal feature interaction between the textual and visual modalities. Subsequently, the visual and textual branches are separately processed by their respective feed-forward networks (FFNs) to further refine the representations. The outputs from both branches are then integrated by a fusion layer, and the resulting multimodal embedding is fed into a three-layer MLP for dimensionality adjustment. This projection step ensures compatibility with the input dimension of SAM’s prompt encoder, enabling the semantic prompt to provide precise and flexible guidance for the downstream anomaly detection task.

This design not only optimizes the fusion of cross-modal features but also enhances the model’s sensitivity to semantic prompts. In this way, the model can dynamically adjust its discrimination ability for anomalous samples based on the textual prompt, effectively improving detection performance.

As shown in Fig. 5, the overall process consists of the following steps: (a) the testing image, (b) the score map gen-

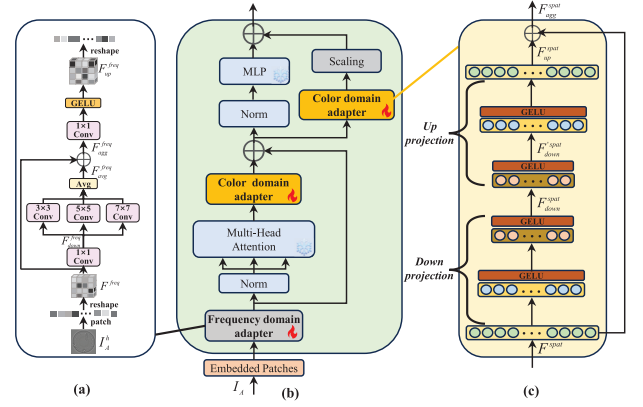


Fig. 6. Dual-branch adapter for SAM. (a) Internal architecture of the frequency-domain adapter. (b) SAM with color- and frequency-domain adapter encoder’s Transformer block. (c) Internal architecture of the color-domain adapter.

erated by the small model, which serves as the initial guidance information, (c) the location and semantic prompts annotated by the multimodal prompt module based on the binarized score map, and (d) the final result of anomaly detection.

D. Dual-Branch Adapter for Sam

Industrial anomaly detection often involves subtle, fine-grained defects and complex textures that differ significantly from the pretrained data distribution of SAM. This domain gap limits the model’s ability to effectively perceive and localize anomalies in real-world industrial settings. Therefore, we propose a dual-branch adapter to enhance the specialization of the SAM-based large model, as illustrated in the blue background at the top right of Fig. 2. The dual-branch adapter parameters are learnable and embedded into the frozen SAM model, enabling effective transfer to industrial scenarios while preserving SAM’s strong general-purpose segmentation capabilities. The adapter consists of two complementary branches: a color-domain adapter and a frequency-domain adapter.

1) *Color-Domain Adapter*: As shown in Fig. 6(c), we design a color-domain adapter with a bottleneck structure. Specifically, the feature embeddings F^{spat} of I_A are first compressed in dimensionality through downprojection to $F^{\text{spat}}_{\text{down}}$, then enhanced in representational capacity through a linear layer to $F^{\text{spat}}_{\text{down}}$, and finally up-projected back to the original dimensionality as $F^{\text{spat}}_{\text{up}}$, which is then residual-connected with F^{spat} to obtain $F^{\text{spat}}_{\text{agg}}$. As shown in Fig. 6(b), the color-domain adapter is integrated into both the main branch and the residual branch of each frozen transformer block. The process can be formalized as follows:

$$F^{\text{spat}}_{\text{down}} = \text{Down}(F^{\text{spat}}) \quad (2a)$$

$$F^{\text{spat}}_{\text{down}} = \text{GeLU}(\text{MLP}(F^{\text{spat}}_{\text{down}})) \quad (2b)$$

$$F^{\text{spat}}_{\text{up}} = \text{Up}(F^{\text{spat}}_{\text{down}}) \quad (2c)$$

$$F^{\text{spat}}_{\text{agg}} = F^{\text{spat}}_{\text{up}} + F^{\text{spat}}. \quad (2d)$$

2) *Frequency-Domain Adapter*: As illustrated in Fig. 6(a), we designed a CNN-based frequency-domain adapter module,

which uses convolutional filters of various sizes to capture complex frequency-domain features at different scales.

As shown in the top-left corner of Fig. 2, the high-frequency component of the image is extracted through frequency-domain filtering. Given a sample I_A , the discrete Fourier transform (DFT) first converts the image to the frequency domain F_A . Then, BHPF is employed to filter out low-frequency components, enhancing details and edges in the image. Finally, the inverse DFT (IDFT) converts the image back to the spatial domain I_A^h

$$F_A = F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} I_A(x, y) e^{-i2\pi(\frac{ux}{M} + \frac{vy}{N})} \quad (3a)$$

$$H_{\text{BHPF}}(u, v) = \frac{1}{1 + (D_0/D(u, v))^{2n}} \quad (3b)$$

$$F(u, v)_A^h = F(u, v) \cdot H_B(u, v) \quad (3c)$$

$$\begin{aligned} I_A^h &= f(x, y)_h \\ &= \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v)_A^h e^{j2\pi(\frac{ux}{M} + \frac{vy}{N})} \end{aligned} \quad (3d)$$

where M and N are the spatial-domain dimensions of the image. u and v correspond to the coordinates in the frequency domain. x and y correspond to the coordinates in the spatial domain.

In order to incorporate high-frequency feature information into the frequency-domain adapter, the high-frequency grayscale image I_A^h is first patched and reshaped into a feature map which is reduced in dimensionality via a 1×1 convolution to obtain $F_{\text{down}}^{\text{freq}}$. Subsequently, $F_{\text{down}}^{\text{freq}}$ undergoes feature extraction through 3×3 , 5×5 , and 7×7 convolutional filters, and the resulting outputs are averaged to produce $F_{\text{avg}}^{\text{freq}}$ which is then combined with $F_{\text{down}}^{\text{freq}}$ through a residual connection to form $F_{\text{agg}}^{\text{freq}}$. Finally, $F_{\text{agg}}^{\text{freq}}$ undergoes dimensional upscaling through a 1×1 convolution and a GeLU activation function, resulting in $F_{\text{up}}^{\text{freq}}$ which is reshaped into a feature embedding. The frequency-domain adapter is inserted before each frozen transformer block, as shown in Fig. 6(b). The process can be formalized as follows:

$$F^{\text{freq}} = \text{Reshape}(\text{Patch}(I_A^h)) \quad (4a)$$

$$F_{\text{down}}^{\text{freq}} = \text{Conv}_{1 \times 1}(F^{\text{freq}}) \quad (4b)$$

$$F_{\text{avg}}^{\text{freq}} = \text{Average} \sum \left(\text{Conv}_{i \times i}(F_{\text{down}}^{\text{freq}}) \right) \quad (4c)$$

$$F_{\text{agg}}^{\text{freq}} = F_{\text{avg}}^{\text{freq}} + F_{\text{down}}^{\text{freq}} \quad (4d)$$

$$F_{\text{up}}^{\text{freq}} = \text{GeLU}(\text{Conv}_{1 \times 1}(F_{\text{agg}}^{\text{freq}})) \quad (4e)$$

The frequency-domain adapter operates on a high-frequency representation of the input image to capture local pattern disruptions. Even slight anomalies, such as microdefects or subtle texture inconsistencies, can produce localized high-frequency responses [37]. To capture both coarse and fine-grained defect patterns, the adapter applies parallel convolutions with multiple kernel sizes. This design enables the model to attend to both large-scale abnormal regions and subtle local defects,

thereby enhancing SAM's ability to perceive anomalies in the frequency domain.

3) *Loss Function*: The dual-branch adapter is trained end-to-end in a self-supervised manner. The loss function is defined as follows:

$$\mathcal{L}_{\text{sam}} = \mathcal{L}_{\text{Dice}}(M_A, M) + \mathcal{L}_{\text{BCE}}(M_A, M) \quad (5a)$$

$$\mathcal{L}_{\text{Dice}}(M_A, M) = 1 - \frac{2M_A M}{M_A + M} \quad (5b)$$

$$\mathcal{L}_{\text{BCE}}(M_A, M) = -(M \log M_A + (1 - M) \log (1 - M_A)) \quad (5c)$$

where $\mathcal{L}_{\text{Dice}}(\cdot)$ is the Dice loss and $\mathcal{L}_{\text{BCE}}(\cdot)$ is the binary cross entropy loss. M is the anomaly mask predicted by SAM which is obtained by fusing the masks corresponding to the location M_{loc} and semantic prompt M_{sem} . M_A is the synthetic known anomaly mask.

E. Optimization of Training Strategy

As shown in Fig. 3, we propose two training optimization strategies to explore the optimal joint training approach for different modules.

1) *Two-Stage Training Strategy*: A two-stage training strategy is proposed to better handle the complexity involved in training a multimodule system. Although this approach helps alleviate issues such as loss balancing and gradient synchronization during the joint optimization of different modules, it adds complexity to the training process.

In the first stage, the small model is trained independently, focusing on generating accurate and informative prompts without being influenced by the complexity of the large model. In the second stage, the adapter component of the SAM-based large model and the multimodal prompt module are jointly trained. This allows the two modules to effectively collaborate in generating and refining location and semantic prompts for anomaly detection.

The loss functions for the two-stage training strategy are given by (1) and (5), respectively, to achieve phase-specific optimization.

2) *Single-Stage Training Strategy*: To simplify the training process, we introduce a straightforward single-stage training strategy, where all modules are jointly trained in a unified training process. This approach simplifies the overall training workflow since all modules are trained together in the same pipeline. The main advantage of this method is that it reduces the complexity of training and allows for a more direct optimization of the entire model.

In this approach, all components of the model, except for the pretrained parameters frozen in the SAM-based large model, are optimized simultaneously. By training all modules together, the model can learn the interdependencies between them, leading to a more consistent and integrated system. This method ensures that the small model, large model, and multimodal prompt module work collaboratively from the beginning of the training process.

The loss function for the entire training process is the sum of the loss functions of the small and large models, which can be expressed as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{small}} + \beta \mathcal{L}_{\text{sam}} \quad (6)$$

TABLE II

ANOMALY DETECTION RESULTS IN TERMS OF AUROC AT IMAGE LEVEL ON THE MVTec AD DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED. THE UNCERTAINTIES ARE QUANTIFIED AS MEAN STANDARD DEVIATION

Category	DSR [28] (ECCV2022)	PatchCore [8] (CVPR2022)	DRAEM [14] (ICCV2021)	RD4AD [7] (CVPR2022)	SimpleNet [15] (CVPR2023)	MDPS [40] (IJCAI2024)	DC-AE [41] (TIJ2024)	AMI-Net [42] (TASE2024)	REASON [43] (TSCVT2025)	DualFlow [44] (TIM2025)	Ours
Carpet	100	99.1	97.0	98.9	99.7	99.6	99.6	<u>99.8</u>	100	100	100 ± 0.00
Grid	100	97.5	<u>99.9</u>	100	99.7	100	100	99.9	100	99.5	100 ± 0.00
Leather	<u>99.3</u>	100	100	100	100	100	100	100	100	100	100 ± 0.00
Tile	100	99.4	99.6	99.3	<u>99.8</u>	100	100	100	<u>99.8</u>	100	100 ± 0.05
Wood	94.7	99.1	99.1	99.2	100	99.1	100	99.3	99.3	100	<u>99.9 ± 0.03</u>
Texture Avg.	98.8	99.1	99.1	99.5	<u>99.8</u>	99.7	99.9	<u>99.8</u>	<u>99.8</u>	99.9	99.9 ± 0.06
Bottle	100	100	99.2	100	100	100	99.8	100	100	100	100 ± 0.00
Cable	93.8	<u>99.5</u>	91.8	95.0	99.9	98.3	95.8	<u>99.5</u>	98.8	99.9	<u>99.3 ± 0.13</u>
Capsule	98.0	97.6	98.5	96.3	97.7	91.4	99.0	98.4	98.5	99.5	<u>99.2 ± 0.22</u>
Hazelnut	95.7	100	100	99.9	100	99.8	100	100	100	100	100 ± 0.00
Metal nut	98.5	<u>99.9</u>	98.7	<u>99.9</u>	100	<u>99.9</u>	100	99.8	100	99.3	100 ± 0.00
Pill	97.5	95.9	98.9	96.6	<u>99.0</u>	96.8	97.8	96.0	98.2	99.1	<u>98.5 ± 0.15</u>
Screw	95.9	94.9	93.9	97.0	98.2	96.7	99.7	97.9	98.5	96.5	<u>99.0 ± 0.08</u>
Toothbrush	100	100	100	99.5	<u>99.7</u>	100	100	96.1	100	96.4	100 ± 0.00
Transistor	97.3	100	93.1	96.7	100	100	96.7	100	99.2	<u>99.6</u>	<u>98.7 ± 0.27</u>
Zipper	100	99.5	100	98.5	<u>99.9</u>	100	100	98.5	99.2	99.5	<u>99.8 ± 0.11</u>
Object Avg.	97.7	98.7	97.4	98.0	99.5	98.3	98.9	98.6	99.2	99.0	<u>99.4 ± 0.18</u>
Average	98.0	98.8	98.0	98.5	99.6	98.8	99.2	99.0	<u>99.4</u>	99.3	99.6 ± 0.15

where β is a hyperparameter that balances the loss weight between the large and small models. The optimal value of β is empirically determined to be 8 based on the ablation study in Section IV-E7.

IV. EXPERIMENTS

In this section, the proposed method is extensively experimented on different datasets to evaluate its effectiveness, and the ablation studies have been experimentally conducted and discussed.

A. Datasets

In our experiment, we primarily use three public unsupervised anomaly detection datasets.

1) *MVTec AD*: The MVTec AD [38] dataset includes ten object categories and five texture categories, with a total of 4096 normal samples and 1258 anomaly samples in the test set. The anomaly types include scratches, dents, and color variations. Pixel-level annotations are provided as the ground truth for evaluation.

2) *WFDD*: The WFDD [16] dataset contains 4101 images of woven fabrics, categorized into four different types with various textures and patterns. The training set consists of 3657 normal samples, whereas the testing set is composed of 203 normal samples and 241 anomaly samples.

3) *BTAD*: The BTAD [39] dataset includes three object categories, with a total of 451 normal images and 290 anomalous images. The images have resolutions ranging from 600 to 1600.

B. Evaluation Metrics

Image-level area under the receiver operating characteristic curve (AUROC) is used to evaluate anomaly detection performance, while pixel-level AUROC and average precision (AP) are employed to assess anomaly localization performance. Notably, due to the significant imbalance between positive and negative samples, AUROC may not effectively reflect localization performance [14]. Therefore, pixel-level AP is

adopted to provide a more accurate evaluation of performance for anomaly localization tasks.

C. Implementation Details

In our experiments, the feature extractor for the small model is WideResNet50. The image encoder of SAM is ViT-B, with its parameters frozen during training. The multimodal model employed is BEiT3-B. The optimizer is Adam, with the learning rate set to 1×10^{-5} . The training is conducted for 200 epochs with a batch size of 8. Our experiments are implemented using the PyTorch framework on a single NVIDIA GeForce RTX 3090 GPU. All images are resized to 224×224 before being input into the model. The model is evaluated under the single-stage training strategy.

D. Experimental Results

1) *Experiments on the MVTec AD Dataset Under Single-Class Setting*: The proposed method is thoroughly compared with state-of-the-art unsupervised anomaly detection methods in both anomaly detection and localization tasks, as shown in Tables II and III. The methods included in the comparison are DSR [28], PatchCore [8], DRAEM [14], RD4AD [7], SimpleNet [15], MDPS [40], DC-AE [41], AMI-Net [42], REASON [43], and DualFlow [44]. To ensure fairness, all results, except those from DC-AE [41], are reproduced using the official code provided in their respective papers. For DC-AE, which does not have the official code, the results are borrowed directly from this article.

The experimental results indicate that the proposed method has a significant advantage in both anomaly detection and localization tasks. Specifically, for anomaly detection, our method achieves an image-level AUROC of 99.6%, demonstrating state-of-the-art performance. In addition, for anomaly localization, our method achieves a pixel-level AUROC of 98.4% and a pixel-level AP of 74.1%, surpassing all the existing state-of-the-art methods. The pixel-level AP is 18.2% higher than that of SimpleNet, showing the superior localization accuracy of our method, particularly in precisely

TABLE III

ANOMALY LOCALIZATION RESULTS IN TERMS OF AUROC/AP AT PIXEL-LEVEL ON THE MVTec AD DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED. THE UNCERTAINTIES ARE QUANTIFIED AS MEAN STANDARD DEVIATION

Category	DSR [28] (ECCV2022)	PatchCore [8] (CVPR2022)	DRAEM [14] (ICCV2021)	RD4AD [7] (CVPR2022)	SimpleNet [15] (CVPR2023)	MDPS [40] (IJCAI2024)	DC-AE [41] (TII2024)	AMI-Net [42] (TASE2024)	REASON [43] (TSCVT2025)	DualFlow [44] (TIM2025)	Ours
Carpet	96.0/78.2	99.1/64.1	95.5/53.5	98.9/56.5	98.2/53.9	94.4/-	97.8/-	99.2/-	99.3/-	98.7/-	98.6 ± 0.08 / 79.3 ± 0.10
Grid	99.6/68.0	98.8/30.9	99.7/65.7	98.3/15.8	98.8/41.4	99.4/-	99.7/-	98.9/-	99.3/-	99.5/-	98.8 ± 0.06 / 60.9 ± 0.12
Leather	99.5/62.5	99.3/45.9	98.6/ <u>75.3</u>	99.4/47.6	99.2/52.3	99.5/-	99.7/-	99.4/-	99.5/-	99.3/-	99.0 ± 0.12 / 76.2 ± 0.21
Tile	98.6/ 93.9	95.7/54.9	99.2/92.3	95.7/54.1	97.0/55.3	96.4/-	98.6/-	96.0/-	96.0/-	96.2/-	98.3 ± 0.12 / 91.4 ± 0.28
Wood	91.5/68.4	95.0/50.0	<u>96.4/77.7</u>	95.8/48.3	94.5/45.5	95.7/-	96.6/-	95.3/-	95.5/-	95.1/-	<u>96.4 ± 0.10</u> / 78.3 ± 0.15
Texture Avg.	97.0/ <u>74.2</u>	97.6/49.2	97.9/72.9	97.6/44.5	97.5/49.8	97.1/-	98.7/-	97.8/-	97.9/-	97.8/-	98.3 ± 0.18 / 76.9 ± 0.25
Bottle	98.8/ 91.5	98.5/77.7	99.1/86.5	98.8/78.0	98.0/75.6	98.6/-	99.4/-	98.8/-	99.0/-	98.3/-	99.0 ± 0.10 / 88.5 ± 0.14
Cable	97.7/ 70.4	98.4/66.3	94.7/52.4	97.2/52.6	97.6/53.2	95.8/-	96.7/-	98.6/-	98.5/-	97.8/-	98.0 ± 0.35 / 64.8 ± 0.58
Capsule	91.0/ 53.3	99.0/44.7	94.3/49.4	98.7/47.2	<u>98.9/51.6</u>	93.6/-	97.2/-	98.9/-	98.7/-	99.0/-	97.9 ± 0.42 / 51.8 ± 0.31
Hazelnut	99.1/87.3	98.7/53.5	99.7/92.9	99.0/60.7	97.9/58.7	98.6/-	99.7/-	98.6/-	99.2/-	98.4/-	<u>99.5 ± 0.05</u> / 88.6 ± 0.14
Metal nut	94.1/67.5	98.3/86.9	99.5/96.3	97.3/78.6	98.8/71.3	97.7/-	<u>99.2/-</u>	96.5/-	97.9/-	97.0/-	<u>99.2 ± 0.61</u> / 87.8 ± 0.85
Pill	94.2/65.7	97.8/ <u>77.9</u>	97.6/48.5	98.1/76.5	98.6/68.2	98.2/-	<u>98.4/-</u>	98.2/-	98.2/-	97.8/-	<u>98.4 ± 0.20</u> / 79.8 ± 0.26
Screw	98.1/52.5	99.5/36.1	97.6/ 58.2	99.7/52.1	99.3/47.7	98.9/-	<u>99.6/-</u>	99.4/-	99.5/-	99.5/-	98.3 ± 0.55 / 52.9 ± 0.78
Toothbrush	99.5/74.2	98.6/38.3	98.1/44.7	99.1/51.1	98.5/49.3	98.8/-	99.3/-	99.9/-	99.1/-	97.5/-	99.1 ± 0.22 / 67.0 ± 0.34
Transistor	80.3/41.1	96.3/66.4	90.9/50.7	92.3/54.1	97.6/46.9	94.7/-	88.8/-	98.2/-	95.4/-	<u>97.8/-</u>	96.6 ± 1.10 / 64.5 ± 1.44
Zipper	98.4/78.5	98.9/62.8	98.8/ 81.5	98.3/57.5	<u>98.9/66.4</u>	98.5/-	99.2/-	98.5/-	98.6/-	<u>98.9/-</u>	98.7 ± 0.24 / 78.8 ± 0.35
Object Avg.	95.1/ <u>68.2</u>	<u>98.4/61.1</u>	97.0/66.1	97.9/60.8	<u>98.4/58.9</u>	97.3/-	97.8/-	98.5/-	98.4/-	98.2/-	98.5 ± 0.95 / 72.7 ± 1.14
Average	95.8/ <u>70.2</u>	98.1/57.1	97.3/68.4	97.8/55.4	98.1/55.9	97.4/-	98.1/-	<u>98.2/-</u>	98.2/-	98.0/-	98.4 ± 0.20 / 74.1 ± 0.35

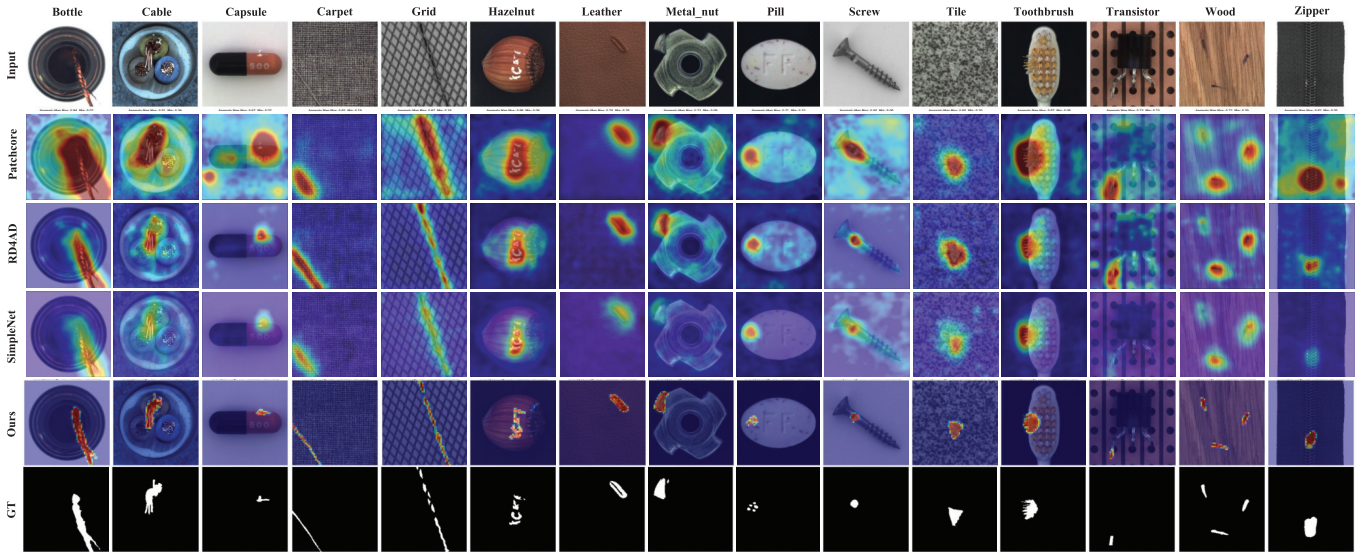


Fig. 7. Qualitative visualization results for anomaly localization on the MVTec AD dataset.

TABLE IV

ANOMALY DETECTION AND LOCALIZATION RESULTS ON MVTec AD DATASET FOR MULTICLASS SETTING. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED. THE UNCERTAINTIES ARE QUANTIFIED AS MEAN STANDARD DEVIATION

Metrics	Padim [9] (ICPR2021)	DRAEM [14] (ICCV2021)	RD4AD [7] (CVPR2022)	UniAD [45] (NIPS2022)	DeSTSeg [46] (CVPR2023)	SimpleNet [15] (CVPR2023)	DiAD [47] (AAAI2024)	Ours
Det.	84.2	88.1	94.6	<u>96.5</u>	89.2	95.3	97.2	97.7 ± 0.15
Loc.	89.5/44.8	88.6/52.6	96.1/48.6	<u>97.0/45.1</u>	93.1/ <u>54.3</u>	96.8/52.6	96.9/45.9	97.5 ± 0.08 / 60.8 ± 0.18

identifying the boundaries of anomalies. This significant improvement suggests that our method is better at capturing the finer details of defects in complex images, thereby enhancing the model's anomaly localization capability.

To visually demonstrate the effectiveness of the proposed method, Fig. 7 presents the visual anomaly score maps of our model on the MVTec AD dataset, along with a comparison to other state-of-the-art methods. These visualizations clearly show that our method outperforms others in both anomaly detection and localization, proving its superiority in capturing fine-grained anomaly details.

2) *Experiments on the MVTec AD Dataset Under Multiclass Setting:* Most existing methods are designed for single-class anomaly detection tasks. However, due to the wide variety of samples to be tested, training a separate model for each class leads to considerable increases in training and memory usage. As a result, multiclass anomaly detection techniques have gradually become a research focus, aiming to train a unified model that can efficiently detect anomalies across all classes, thus reducing the computational and memory overhead.

As shown in Table IV, we compared our proposed LScAD method with several existing multiclass anomaly detection

TABLE V
ANOMALY DETECTION (IMAGE AUROC) AND LOCALIZATION (PIXEL AUROC/PIXEL AP) RESULTS ON WFDD AND BTAD DATASETS

Methods	WFDD		BTAD	
	Det.	Loc.	Det.	Loc.
DSR [28]	95.6	91.7/36.8	92.7	92.7/42.9
PatchCore [8]	96.3	98.6/50.4	95.5	94.5/51.8
DRAEM [14]	91.8	94.4/42.5	91.7	91.7/36.5
RD4AD [7]	94.8	97.4/46.5	94.8	93.8/45.3
SimpleNet [15]	<u>98.2</u>	<u>98.2/54.8</u>	<u>96.4</u>	<u>96.2/48.8</u>
Ours	98.3	98.4/66.4	96.7	96.6/60.9

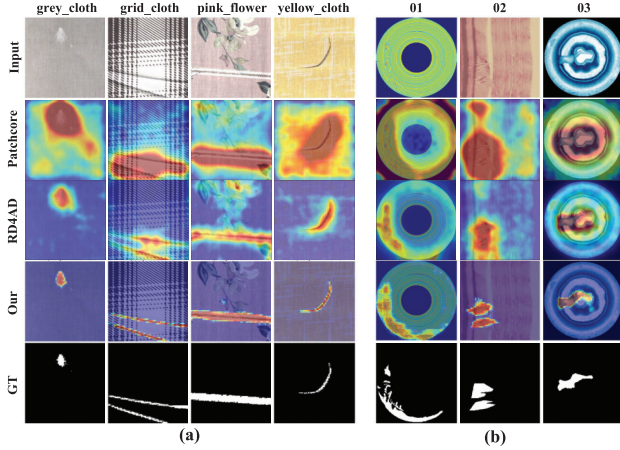


Fig. 8. Qualitative visualization results for anomaly localization on (a) WFDD and (b) BTAD datasets.

methods, including Padim [9], DRAEM [14], RD4AD [7], UniAD [45], DeSTSeg [46], SimpNet [15], and DiAD [47]. The experimental results show that LScAD outperforms these methods, especially in terms of pixel-level AP, demonstrating the superiority of our approach. It is noteworthy that although UniAD [45] and DiAD [47] are specifically designed for multiclass anomaly detection, our method not only achieves state-of-the-art performance in single-class settings but also surpasses these specialized algorithms in multiclass settings. This indicates that our model has strong flexibility, easily transitioning from single-class to multiclass tasks while maintaining high efficiency across different settings.

3) *Experiments on WFDD and BTAD Datasets:* Table V presents the results of anomaly detection and localization for the proposed method and several state-of-the-art unsupervised anomaly detection methods on the WFDD and BTAD datasets. On the BTAD dataset, our method achieves comprehensive state-of-the-art performance in both detection and localization metrics, with a 9.1% improvement in pixel-level AP. On the WFDD dataset, which contains anomalies primarily in the form of points or lines with small areas, our method achieves state-of-the-art performance in both image-level AUROC and pixel-level AP. The anomalies in the WFDD dataset are typically subtle and low contrast, making detection and localization particularly challenging. Despite this, our method performs excellently in capturing these defects, demonstrating its robustness in handling subtle, low-contrast anomalies.

Fig. 8 presents the visual anomaly score maps of our method on both WFDD and BTAD datasets, along with a comparison

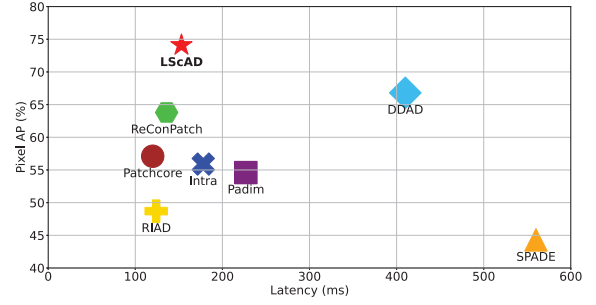


Fig. 9. Comparisons of different anomaly detection methods in terms of pixel AP (y-axis) and latency (x-axis).

TABLE VI
PARAMETER AND INFERENCE SPEED OF SMALL AND LARGE MODELS

Model	Parameter (M)	Inference Speed (ms)
Small Model	~60	~20
Large Model	~650	~130

TABLE VII
QUANTITATIVE RESULTS ON REAL-WORLD DATA

Methods	Det.	Loc.
Patchcore [8]	98.6	<u>98.2/48.9</u>
DRAEM [14]	97.3	96.7/41.3
RD4AD [7]	<u>98.9</u>	98.1/58.4
SimpleNet [15]	98.5	97.2/51.8
Ours	99.2	98.6/66.1

to other methods. These visualizations highlight our model's effective capability to detect and localize anomalies. Intuitively, our method excels in anomaly localization for subtle, texture-based defects, which are typically difficult for other models to accurately capture. This success can be attributed to the collaborative working mode of the small and large models in LScAD. The small model acts as a lightweight guidance information generator, while the large model refines the details and contextual understanding of the anomalies, thus improving the accuracy of anomaly localization. This collaborative approach enables the model to focus on both fine-grained details and broader patterns, making it highly effective across a variety of tasks.

4) *Execution Time Evaluation:* We conducted a real-time performance evaluation of our LScAD on the MVTec AD dataset and compared it with other methods. The inference speed, as presented in Fig. 9, was measured on an NVIDIA RTX 3090 GPU. The experimental results show that our method achieves a latency of 153 ms while delivering the best performance in pixel-level metrics. Furthermore, the parameter size and inference speed of the small and large models in our method are shown in Table VI. The small model has less than 1/10 the number of parameters of the large model.

5) *Applications in Real-World System and Data:* Fig. 10(b) presents the visualization results of anomaly detection and localization, which demonstrate that our method produces accurate and fine-grained segmentation masks for defect regions. In addition, Table VII shows that our method achieves superior performance in both image-level AUROC

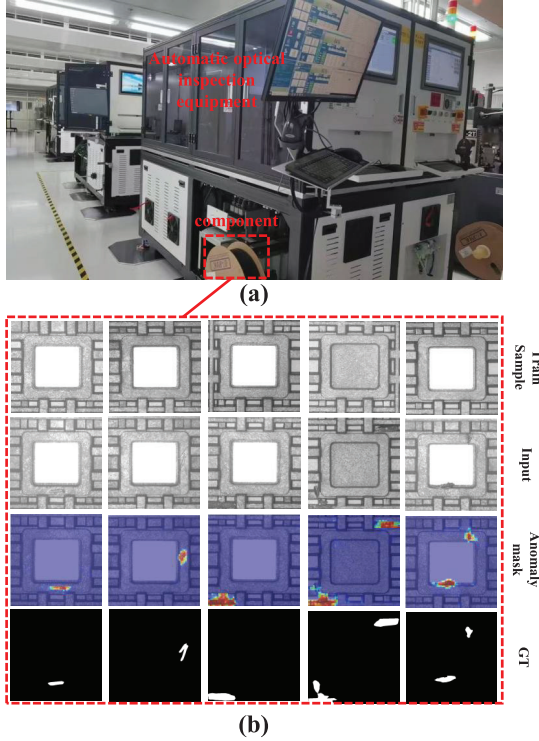


Fig. 10. Applications in real-world systems. (a) Real photograph of the vision hardware system. (b) Qualitative visualization results for anomaly localization on the real-world chip tray dataset.

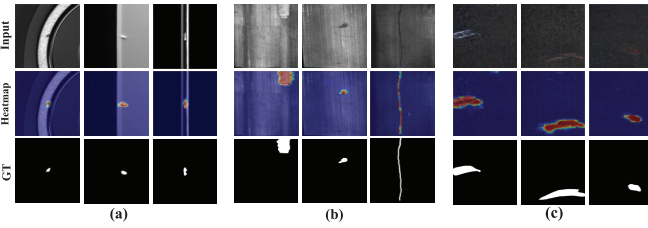


Fig. 11. Three real-world industrial datasets. (a) Smartphone back cover glass dataset. (b) Magnetic Tile. (c) KSDD2.

and pixel-level AP compared to existing approaches. The improvement in AP reflects more precise localization of subtle defects, which is particularly important in real-world industrial scenarios. These results confirm the practical effectiveness and robustness of our method under actual imaging conditions.

To further validate the robustness and credibility of our method across diverse and challenging real-world industrial scenarios, we conducted experiments on three representative datasets, as illustrated in Fig. 11. (a) a smartphone back cover glass dataset from the actual production line, and (b) and (c) two public industrial benchmarks, Magnetic Tile [48] and KSDD2 [49]. These three datasets exhibit varying lighting conditions and complex backgrounds, demonstrating that our method can effectively adapt to different inspection scenarios.

6) *Noise Robustness Analysis*: To evaluate the robustness of the proposed method under degraded visual conditions, we conducted experiments by adding different levels of salt-and-pepper noise (5%, 25%, and 50%) to the original input image. As illustrated in Fig. 12, the localization results remain relatively accurate when the noise level is low (5%), with the anomaly region clearly highlighted in the heatmap. When

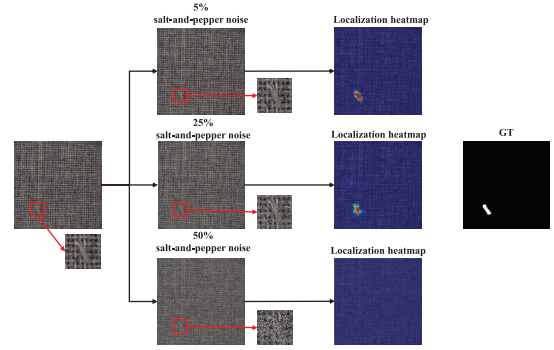


Fig. 12. Noise robustness analysis.

TABLE VIII
INFLUENCE OF THE TRAINING STRATEGY

Training strategy	Det.	Loc.
Single stage	99.6	98.4/74.1
Two stage	99.5	98.6 /73.6

TABLE IX
INFLUENCE OF THE SAM ADAPTER

color domain	frequency domain	Det.	Loc.
×	×	83.5	86.4/37.8
✓	×	98.4	98.6 /70.6
×	✓	94.2	93.5/48.7
✓	✓	99.6	98.4 /74.1

the noise level increases to 25%, the segmentation quality begins to degrade, but the model can still approximately identify the defective region. However, at 50% noise, the anomaly region becomes nearly indistinguishable even to the human eye, as shown in the zoomed-in patch. Therefore, the model fails to respond to the anomaly region, leading to missed detection. These results suggest that our method maintains reasonable robustness under moderate noise, but its performance can be compromised when the defect itself is obscured. This highlights the necessity of applying denoising or image enhancement preprocessing in extremely noisy industrial environments.

E. Ablation Study

In this section, we conduct extensive ablation experiments and analysis of the proposed method on the MVTEC AD dataset.

1) *Influence of the Training Strategy*: We have analyzed the performance of LScAD under different training strategies with the corresponding experimental results presented in Table VIII. The results show that both strategies yield comparable performance in terms of anomaly detection and localization. This suggests that the proposed framework is not sensitive to the training pipeline design, demonstrating its flexibility and ease of deployment. Based on these findings, we adopt the simpler single-stage strategy to reduce training complexity without sacrificing performance.

2) *Influence of the Sam Adapter*: We investigated the impact of the components of the SAM adapter on the proposed method's performance using the MVTEC AD dataset,

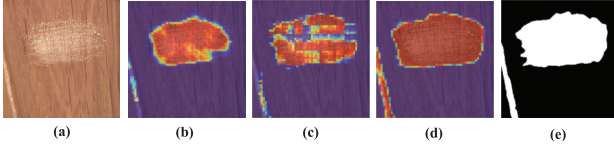


Fig. 13. Visualization results using different forms of prompts. (a) Input image. (b) Only location prompts. (c) Only semantic prompts. (d) Both location prompts and semantic prompts. (e) GT.

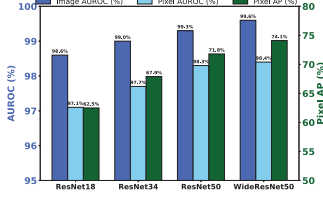


Fig. 14. Influence of the backbone of the small model.

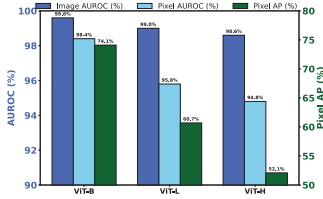


Fig. 15. Influence of the backbone of the large model.

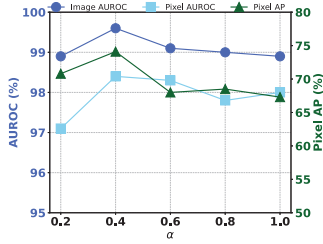


Fig. 16. Influence of the hyperparameter α .

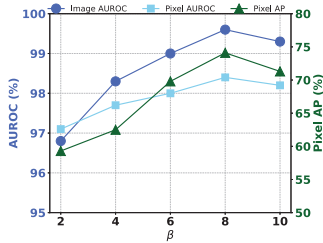


Fig. 17. Influence of the hyperparameter β .

with detailed results presented in Table IX. The experimental results show that without the adapter, the model struggles to generalize to the industrial domain, achieving only 37.8% in pixel-level AP. The addition of the color-domain adapter leads to a significant improvement of 32.8%, indicating its effectiveness in capturing global appearance information. Further introducing the frequency-domain adapter brings an additional 10.9% gain, suggesting its ability to capture fine-grained local anomalies through multiscale frequency features. When both adapters are combined, the model achieves a pixel-level AP of 74.1%, confirming that the dual-branch adapter design provides complementary benefits and plays a critical role in enhancing SAM's adaptation to industrial anomaly detection.

TABLE X
INFLUENCE OF PROMPT FORMATS

point	box	text	Det.	Loc.
✓	✗	✗	98.6	97.7/64.9
✗	✓	✗	98.2	96.7/60.1
✗	✗	✓	98.0	96.3/56.3
✓	✓	✗	99.3	98.6/70.5
✓	✓	✓	99.6	98.4/74.1

TABLE XI
INFLUENCE OF SAM TRAINABLE COMPONENTS

mask decoder	prompt encoder	Det.	Loc.
✗	✗	97.5	95.7/53.8
✓	✗	99.2	97.8/70.2
✗	✓	99.0	97.3/66.1
✓	✓	99.6	98.4/74.1

3) *Influence of the Prompt Formats*: To validate the effectiveness of the multimodal prompt module, we conducted a series of comparative experiments, with quantitative results presented in Table X. We also visualized the anomaly localization outputs using heatmaps under different prompt configurations, as shown in Fig. 13. The results show that the inclusion of the multimodal prompt leads to more accurate and complete localization of anomalies, with significantly fewer false positives. Furthermore, the consistency between the quantitative improvements and visual results confirms that the multimodal prompt module effectively enhances both anomaly detection and localization accuracy in practical scenarios.

4) *Influence of the Sam Trainable Components*: We conducted a comparative experiment to explore the effect of making SAM's prompt encoder and mask decoder either trainable or frozen. As shown in Table XI, the experimental results demonstrate that enabling training for both components significantly improves the model's performance in anomaly detection and localization. This improvement suggests that fine-tuning these modules allows the model to better adapt SAM's pretrained representations to the specific characteristics of industrial anomaly data. In particular, the trainable configuration enhances the model's ability to align prompt information with downstream segmentation tasks, resulting in more accurate and robust localization of anomalies.

5) *Influence of the Backbone of Small and Large Models*: Fig. 14 displays the qualitative comparisons of different backbones of the small model. Intuitively, the deeper and wider WideResNet50 enables the small model to generate high-quality guidance information, which contribute to more accurate anomaly detection. Fig. 15 presents a comparison of our method across different image encoder backbones of SAM. Notably, our method achieves the best performance with the lightweight ViT-B backbone. This is attributed to the fact that smaller backbones are better suited for fine-tuning tasks on the data-limited datasets.

6) *Influence of the Text Prompt Template*: As shown in Table XII, we evaluate the impact of various text prompt templates on the performance of the proposed LScAD frame-

TABLE XII
INFLUENCE OF THE TEXT PROMPT TEMPLATE

text prompt	Det.	Loc.
'a good / defective [']	94.8	95.4/ 56.3
'a photo of a good / defect [']	95.5	95.7/ 60.2
'good / defect'	99.6	98.4/ 74.1
'good / damaged'	99.2	98.5 /72.5
'good / broken'	99.4	98.3/72.8
'normal / abnormal'	98.5	97.8/66.3
'perfect / flawed'	97.9	97.0/64.6

work. Specifically, we compare concise class-level prompts, such as “good/defect” with more descriptive formulations inspired by CLIP-style templates, including “a good/defective [’],” and “a photograph of a good/defect [’].” The experimental results show that concise prompt words generally outperform CLIP-style text prompt templates. Among them, the prompt “good/defect” achieves the best overall performance, suggesting that in industrial anomaly detection tasks, minimal semantic cues are sufficient to effectively guide the multimodal encoder. Moreover, this design simplifies the prompt construction process and enhances the generalizability of the proposed method.

7) *Influence of the Hyperparameter of Loss Function:* Under the single-stage training strategy, we investigated the impact of two key hyperparameters, α and β , on the performance of our model. Specifically, α controls the contribution of the reconstruction loss $\mathcal{L}_{\text{Rec}}$ relative to the knowledge distillation loss $\mathcal{L}_{\text{kd}}$ in the small model, while β determines the weight of the segmentation loss $\mathcal{L}_{\text{sam}}$ derived from the large SAM-based model in the overall objective. We conducted multiple comparative experiments with different settings of these hyperparameters, and the results are shown in Figs. 16 and 17. The experimental findings indicate that setting $\alpha = 0.4$ and $\beta = 8$ yields the best performance across various anomaly detection and localization metrics, suggesting that a well-balanced tradeoff between the small- and large-model objectives is crucial for optimal performance.

V. CONCLUSION

In this work, we propose LScAD, a novel framework for unsupervised industrial anomaly detection that leverages the collaboration between small and large models. This framework effectively combines the task-specific orientation of small models with the generalization capability of large models, achieving complementary strengths. Specifically, the small model is utilized to generate the initial guidance information for the large model, overcoming the limitations of SAM’s interactive working mode. The multimodal prompt module supplies SAM with diverse and rich prompt information for subsequent anomaly segmentation tasks. Moreover, the dual-branch adapter mitigates biases in SAM’s pretrained data distribution, effectively transferring its segmentation capabilities to industrial anomaly detection scenarios. Extensive experiments on the MVTec AD dataset and other real-world datasets demonstrate the effectiveness of the proposed method.

Meanwhile, we also acknowledge the current limitations, due to the integration of the large model-SAM, the overall parameter count and inference speed of the model still

have room for improvement. In the future, we will further enhance the model’s real-time performance and computational efficiency through techniques such as model distillation and pruning.

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